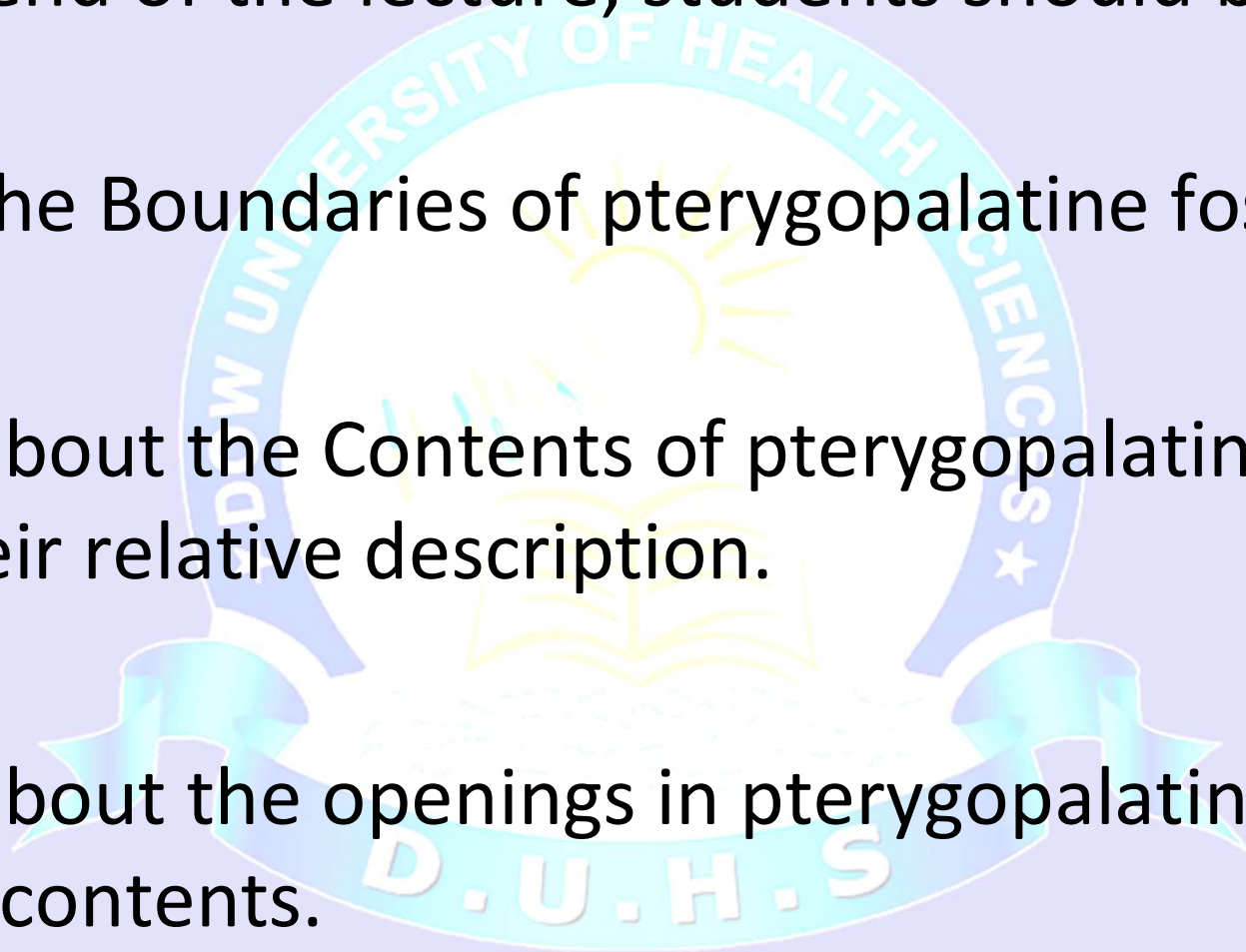




PTERYGOPALATINE **FOSSA.**

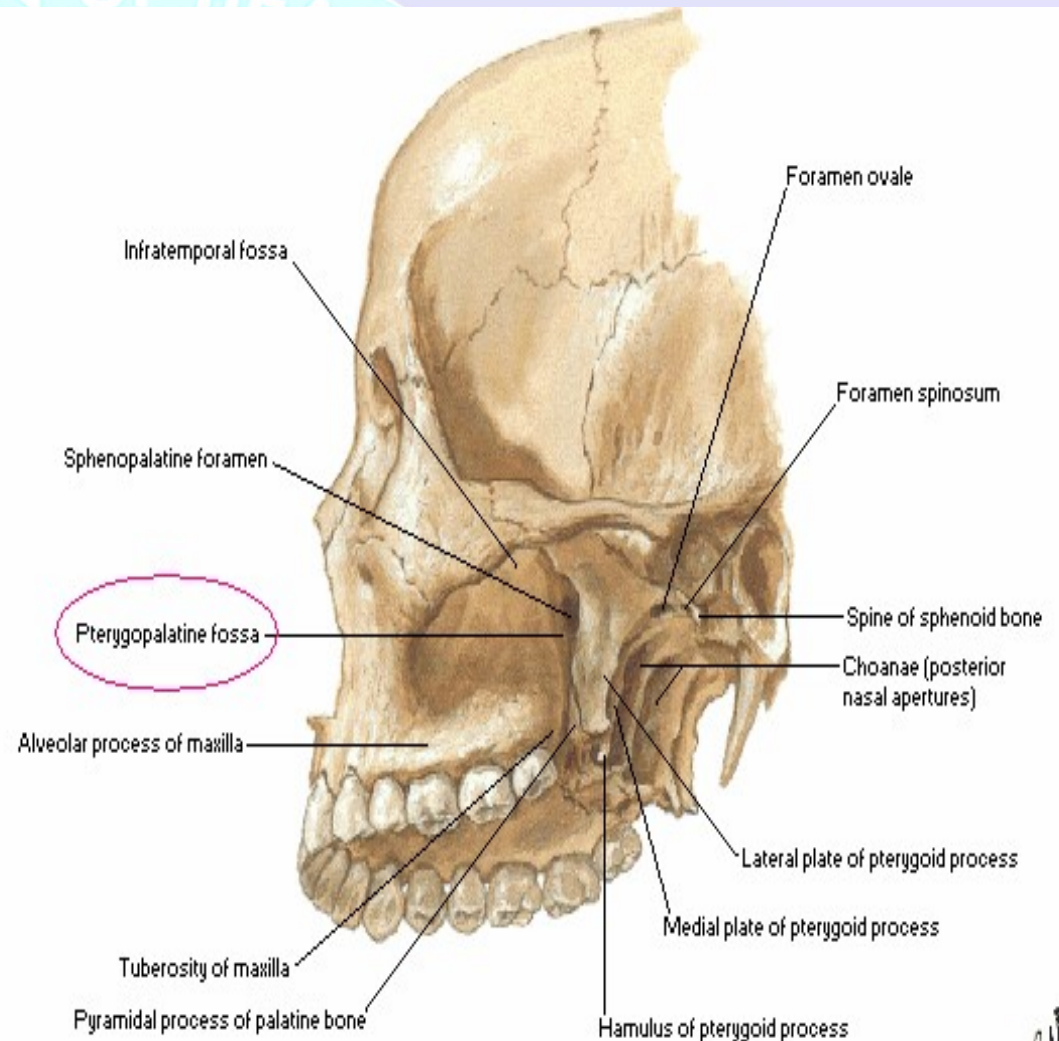
Learning Objectives.

- At the end of the lecture, students should be able to:
- Know the Boundaries of pterygopalatine fossa.
- Know about the Contents of pterygopalatine fossa and their relative description.
- Know about the openings in pterygopalatine fossa, and its contents.



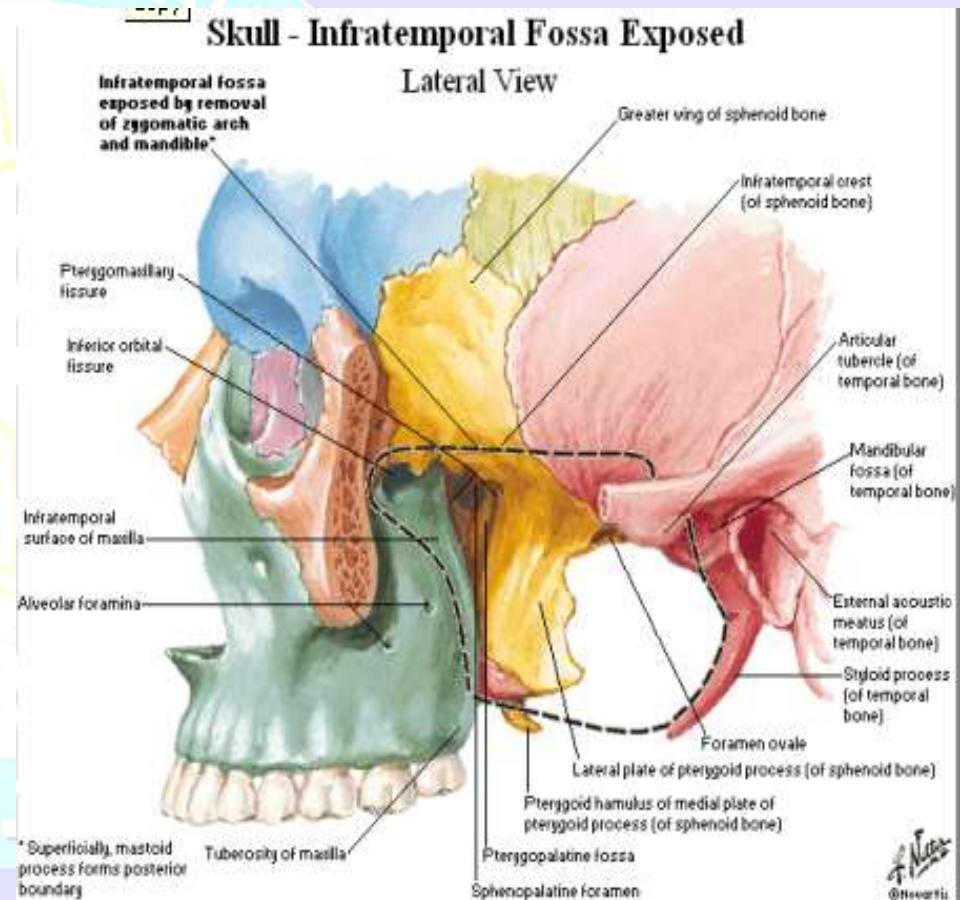
PTERYGOPALATINE FOSSA.

- It is a small space between the back of the maxilla and the front of the pterygoid process of the sphenoid.
- Concerned with the blood and nerve supply of the upper jaw.



BOUNDARIES.

- Anterior -
superomedial part of the infratemporal surface of maxilla.
- Posteriorly -
Sphenoid (root of the pterygoid process)
- Laterally -
Pterygomaxillary fissure (between maxilla & lateral pterygoid plates).
- Medially- Palatine bone
- Roof -
Orbital process of palatine bone



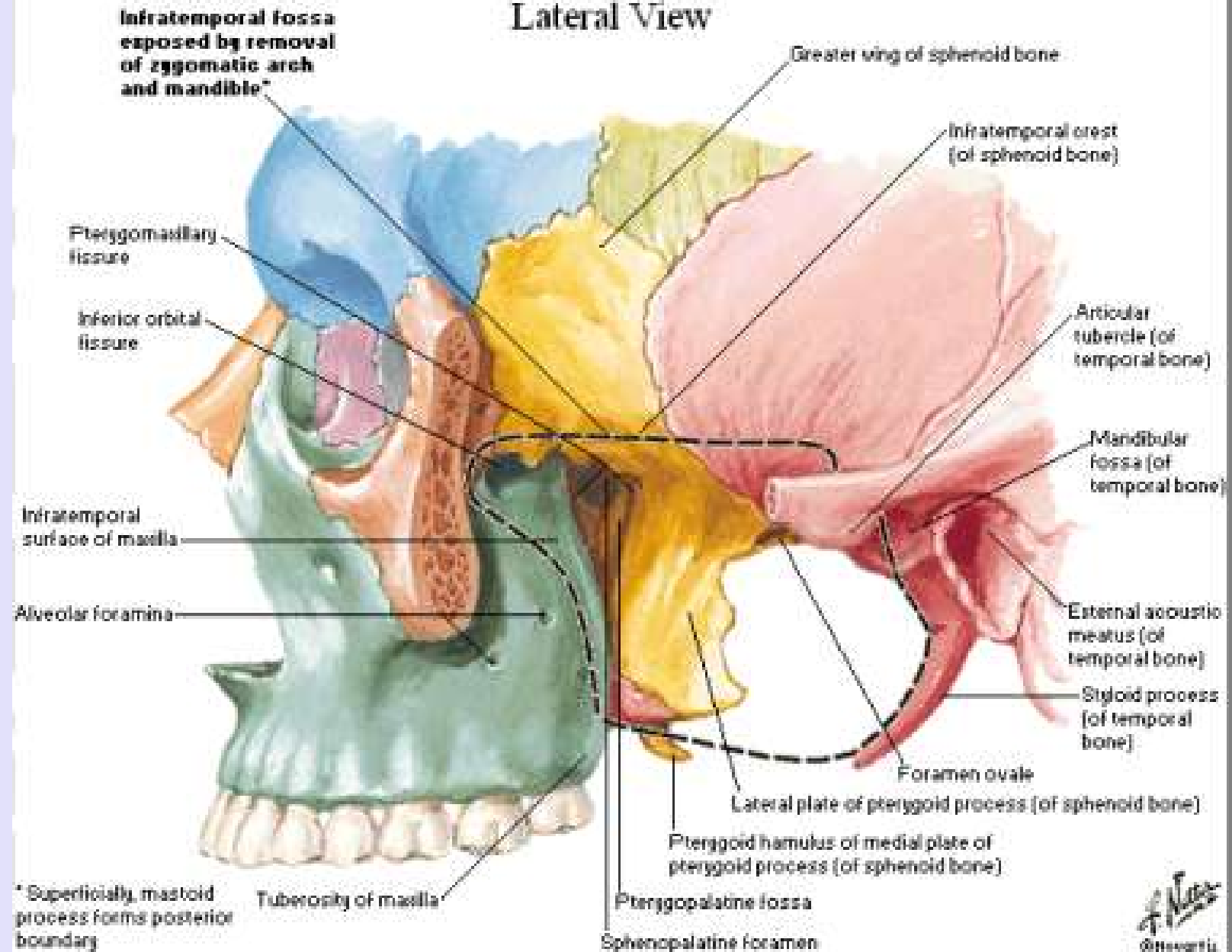
BOUNDARIES OF PTERYGOPALATINE FOSSA.

(DETAILS).

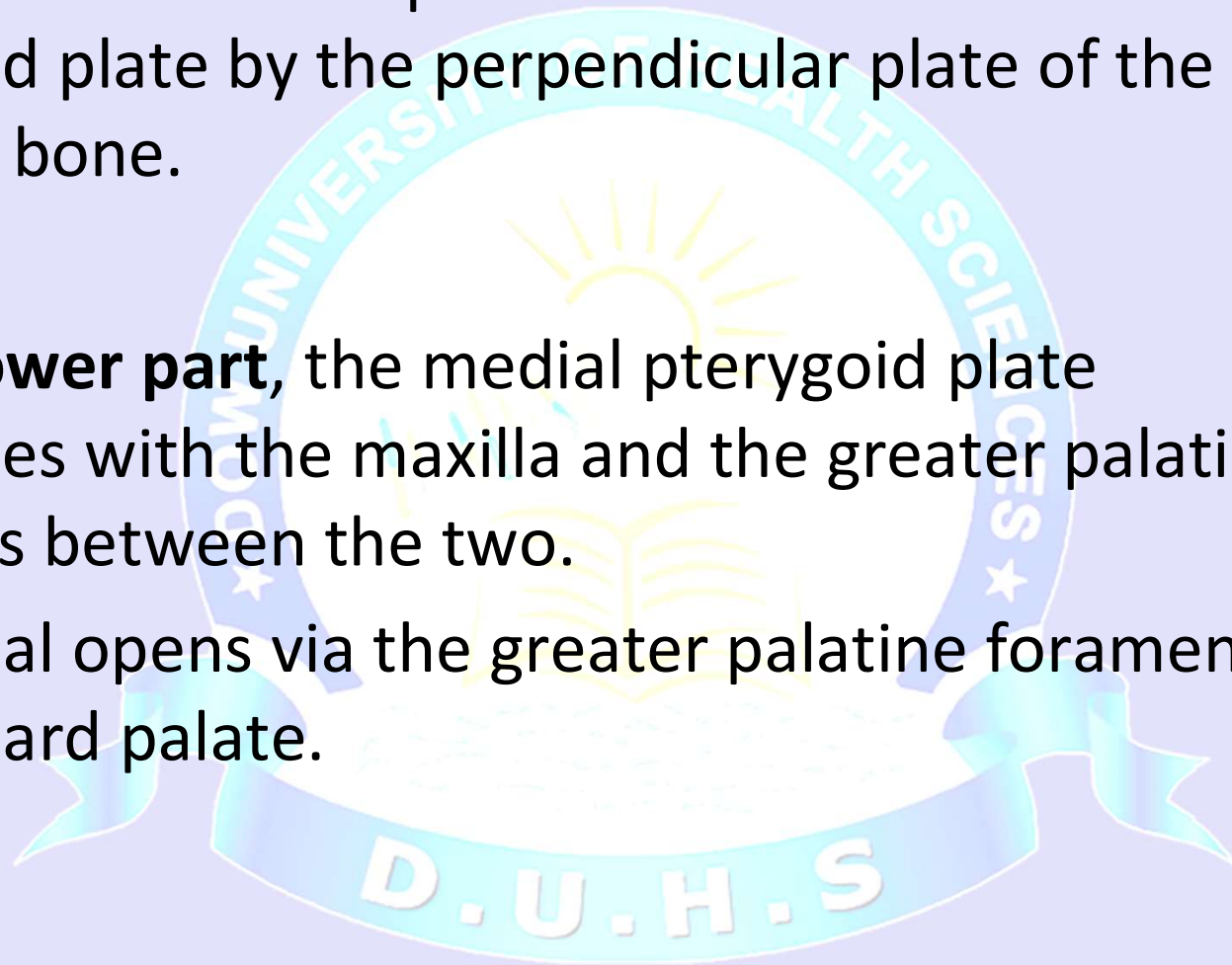
- **Anteriorly** is the posterior wall of the maxilla.
- **Posteriorly** is the sphenoid bone. Here the root of the pterygoid process contains the pterygoid canal and the greater wing of the sphenoid contains the foramen rotundum.
- **Laterally** is the pterygomaxillary fissure between the maxilla and the lateral pterygoid plate. The fissure is closed inferiorly where the maxilla and the lateral pterygoid plate are joined by the pyramidal process of the palatine bone.

Skull - Infratemporal Fossa Exposed

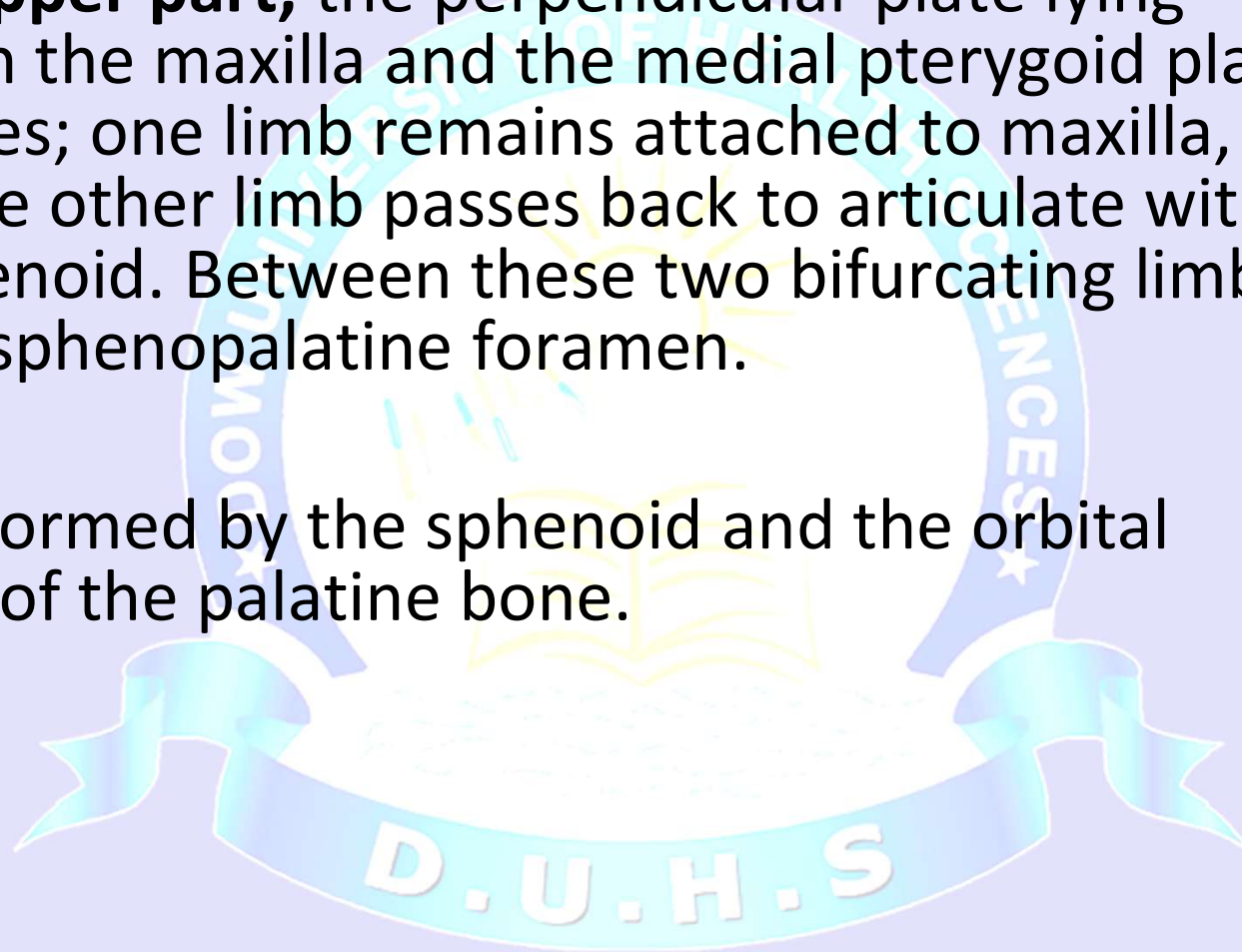
Lateral View



- **Medially** is the lateral wall of the nose.
- Here the maxilla is separated from the medial pterygoid plate by the perpendicular plate of the palatine bone.
- **In the lower part**, the medial pterygoid plate articulates with the maxilla and the greater palatine canal lies between the two.
- The canal opens via the greater palatine foramen on the hard palate.

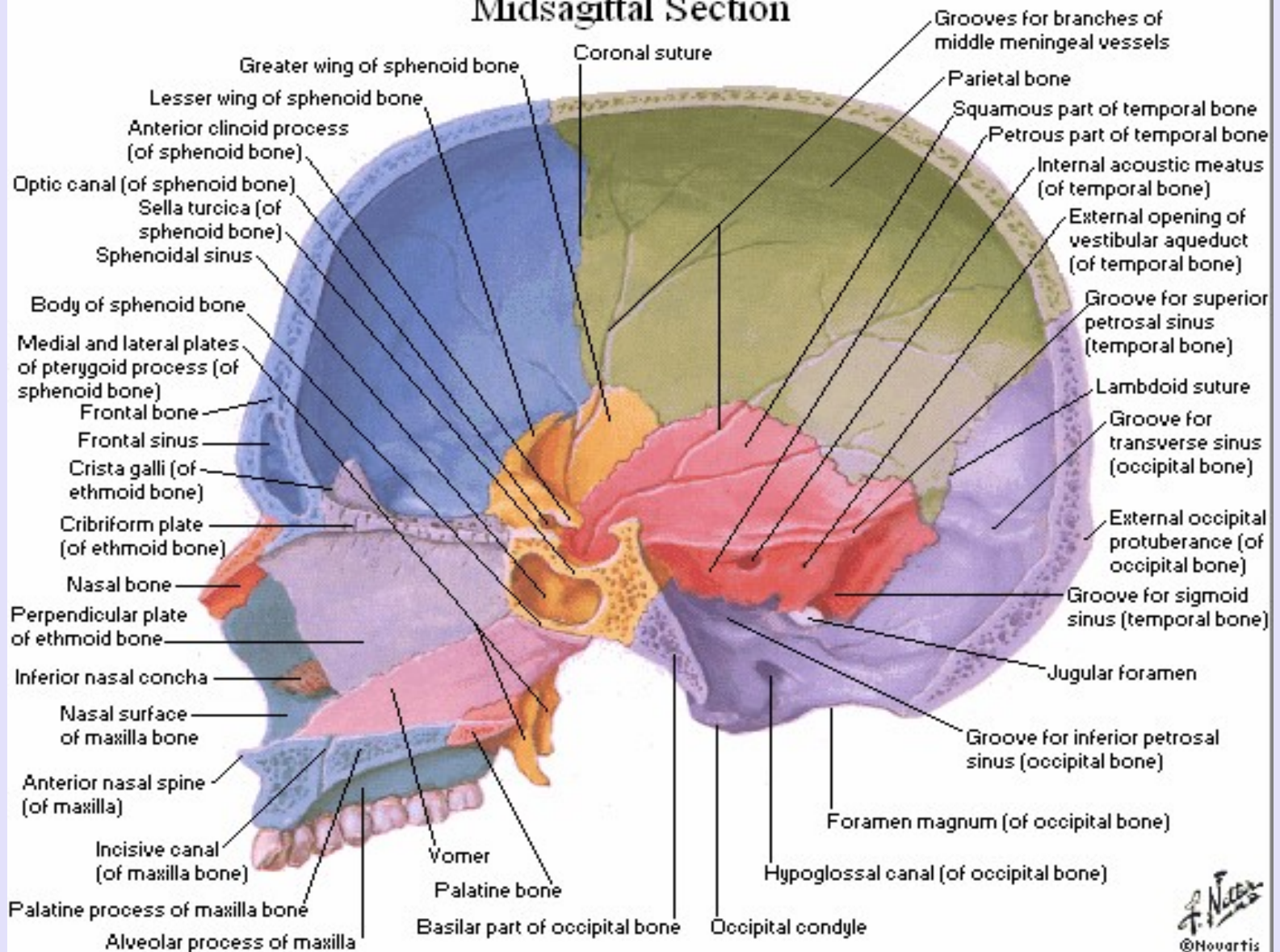


- **In the upper part**, the perpendicular plate lying between the maxilla and the medial pterygoid plate bifurcates; one limb remains attached to maxilla, while the other limb passes back to articulate with the sphenoid. Between these two bifurcating limbs lies the sphenopalatine foramen.
- **Roof** is formed by the sphenoid and the orbital process of the palatine bone.

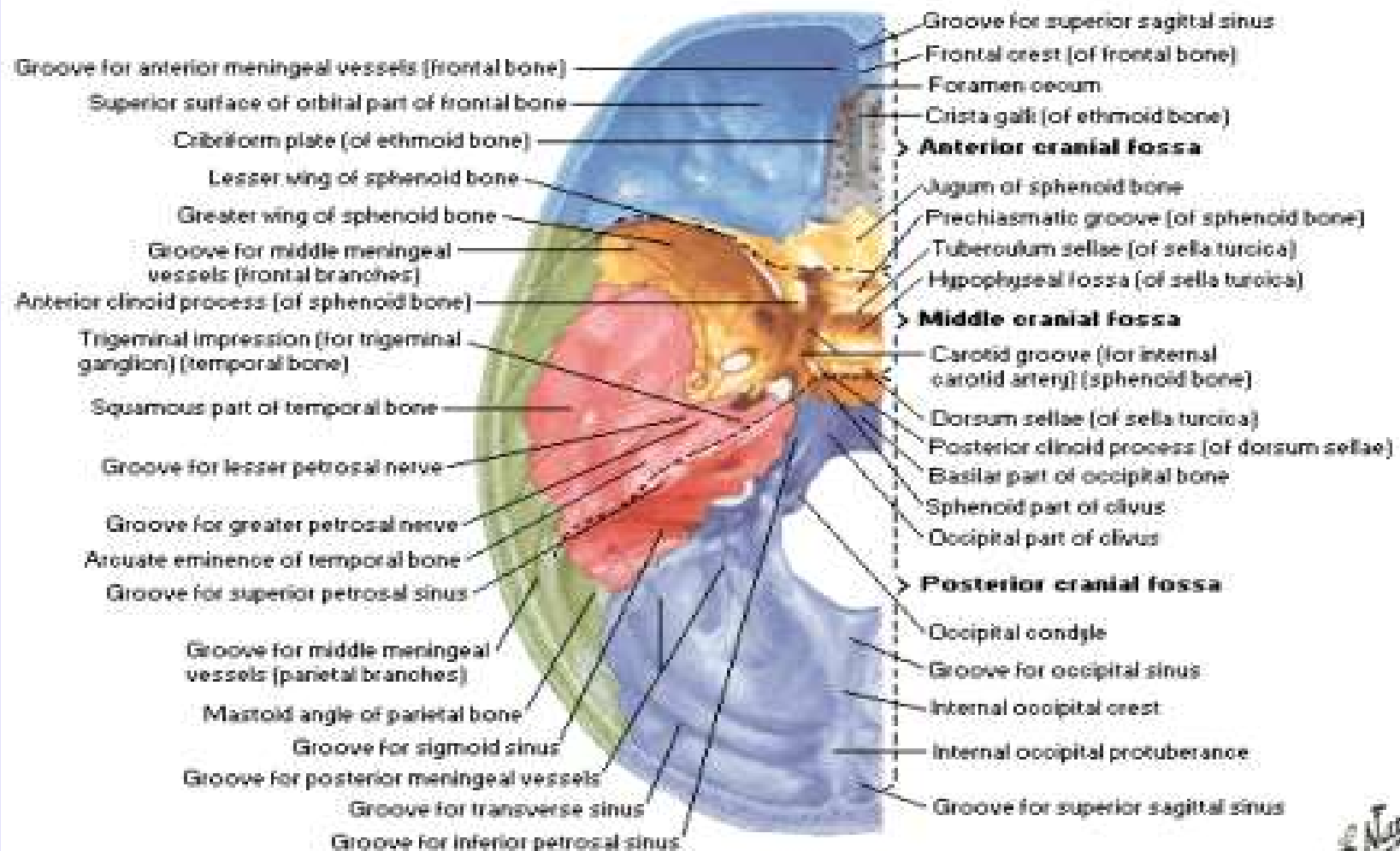


Skull

Midsagittal Section



Bones of Cranial Base Superior View



Contents of pterygopalatine fossa.

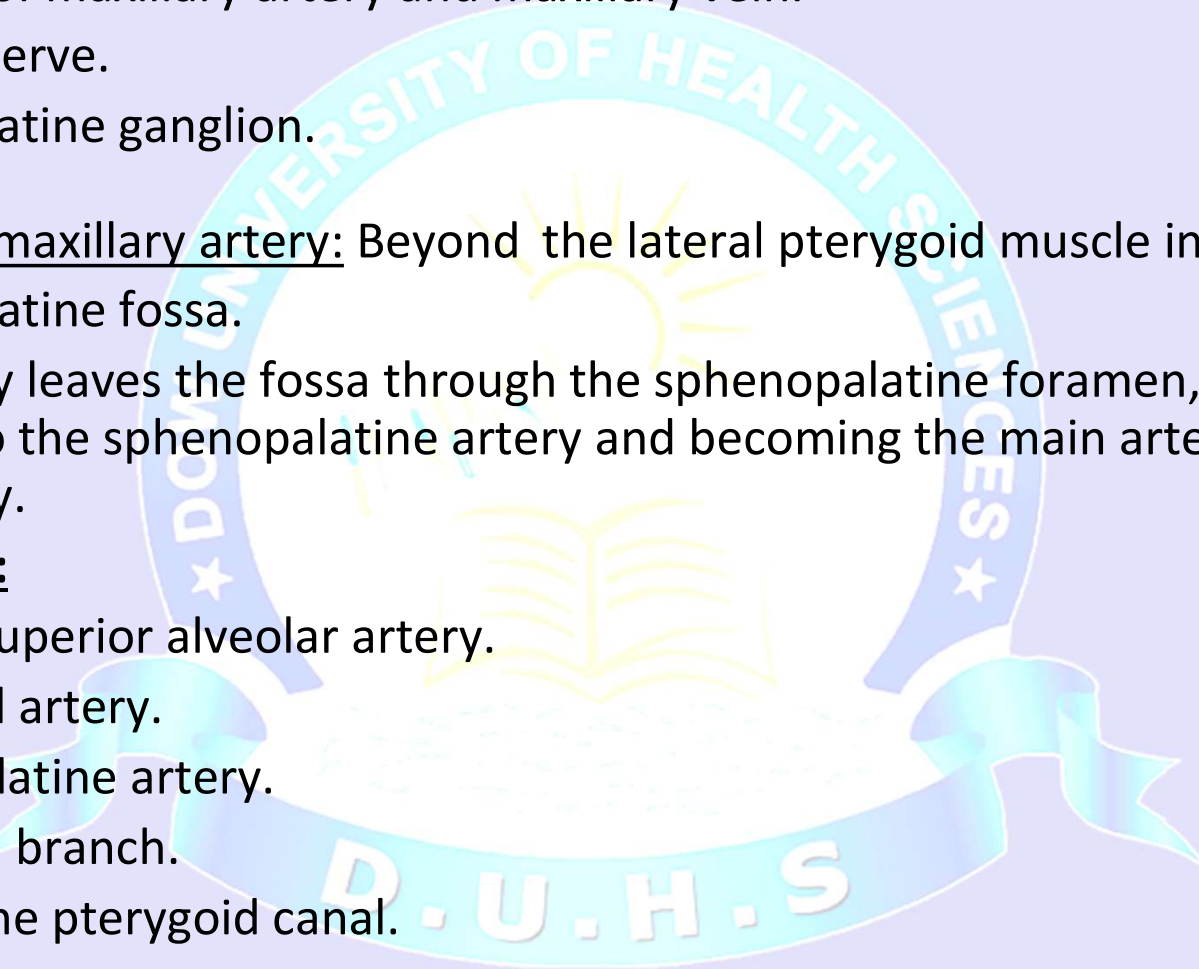
1. Third part of maxillary artery and maxillary vein.
2. Maxillary nerve.
3. Pterygopalatine ganglion.

1. Third part of maxillary artery: Beyond the lateral pterygoid muscle into the pterygopalatine fossa.

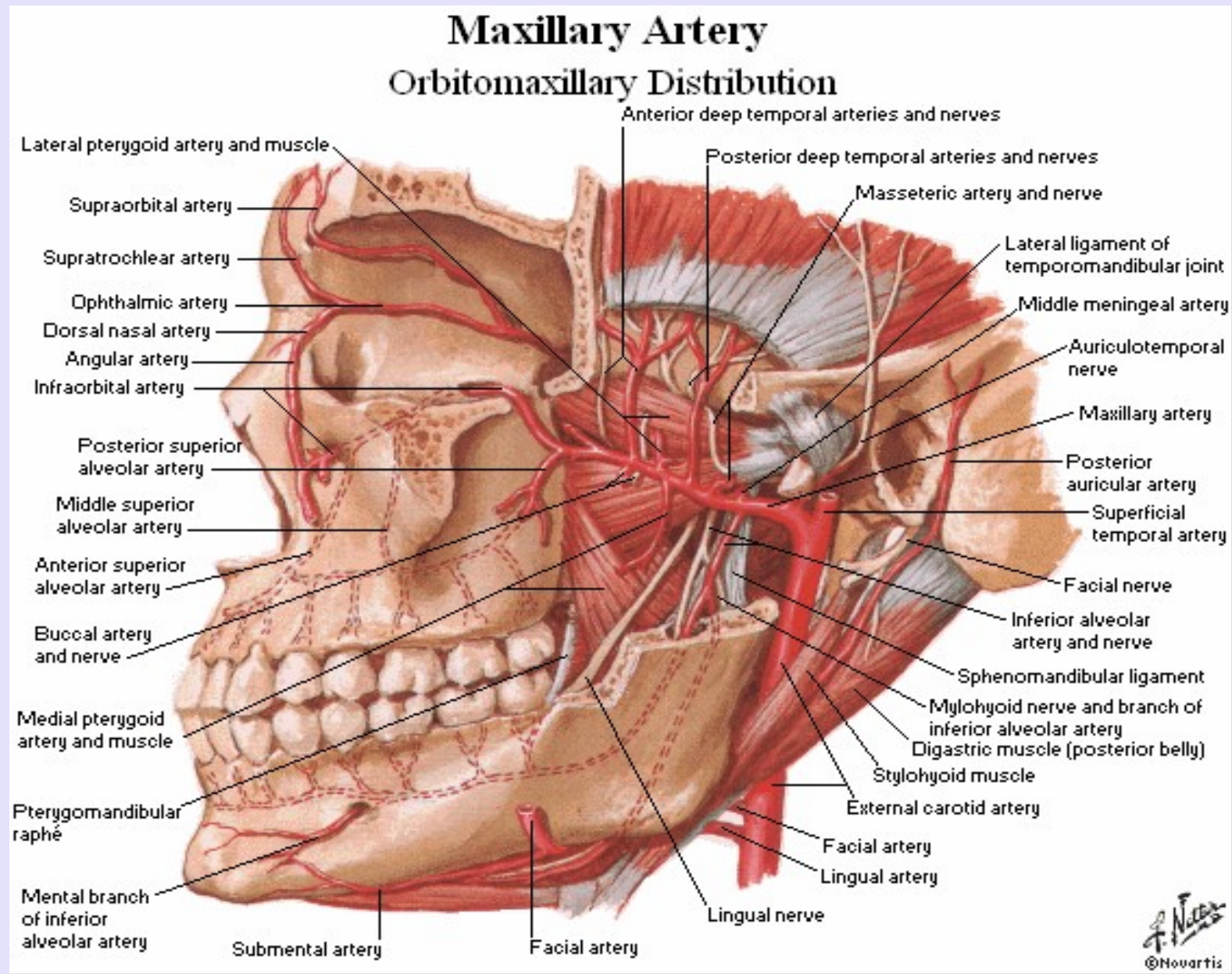
- The artery leaves the fossa through the sphenopalatine foramen, changing its name to the sphenopalatine artery and becoming the main artery of the nasal cavity.

- **Branches:**

- Posterior superior alveolar artery.
- Infraorbital artery.
- Greater palatine artery.
- Pharyngeal branch.
- Artery of the pterygoid canal.
- Sphenopalatine artery.



1. Maxillary artery.



Artery of pterygoid canal

Sphenopalatine artery

Maxillary artery

Posterior superior alveolar artery

Infraorbital artery

(A)

Pharyngeal artery

Posterior nasal septal artery

Posterior nasal lateral artery

Descending palatine artery

Lacrimal nerve and gland

Zygomatico-temporal and zygomatico-facial nerves

Maxillary nerve

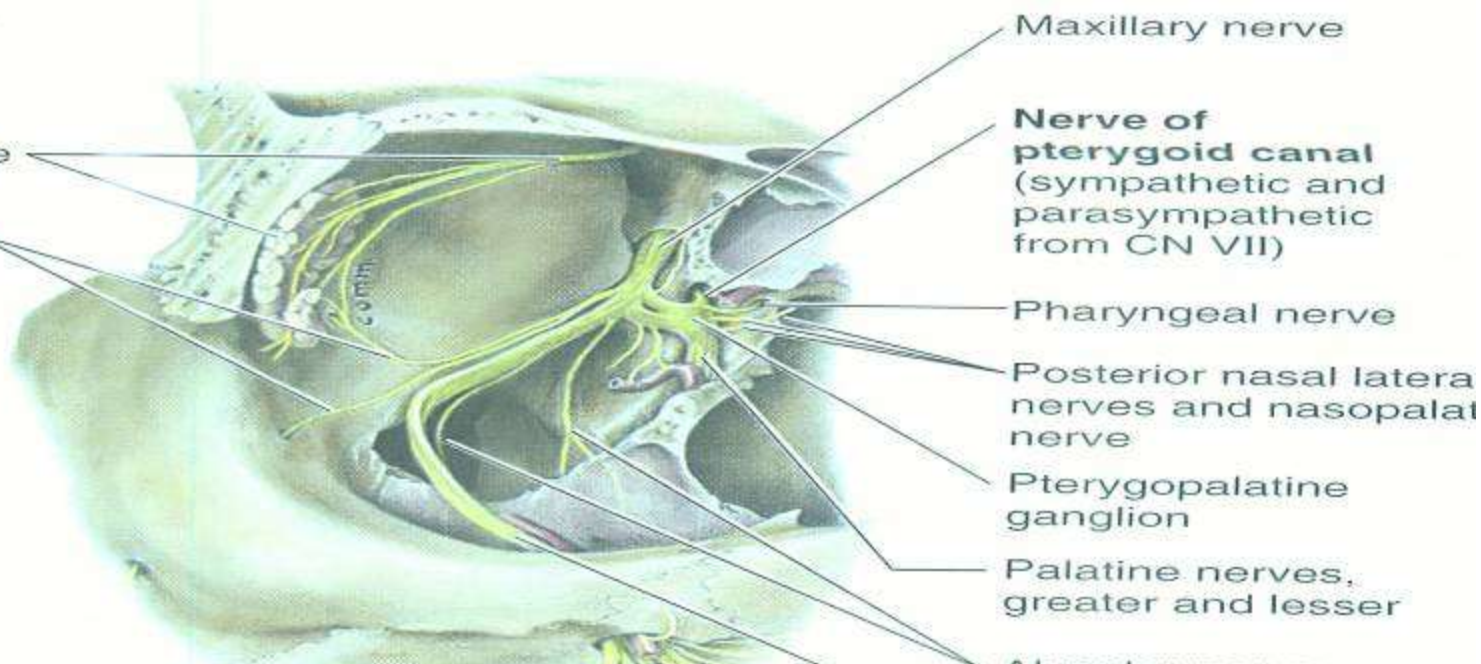
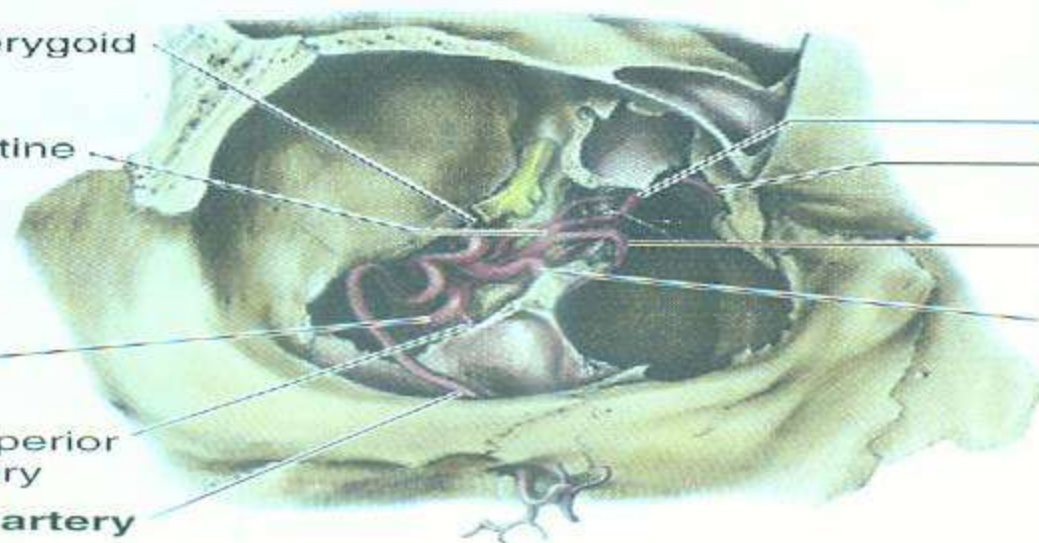
Nerve of pterygoid canal
(sympathetic and parasympathetic from CN VII)

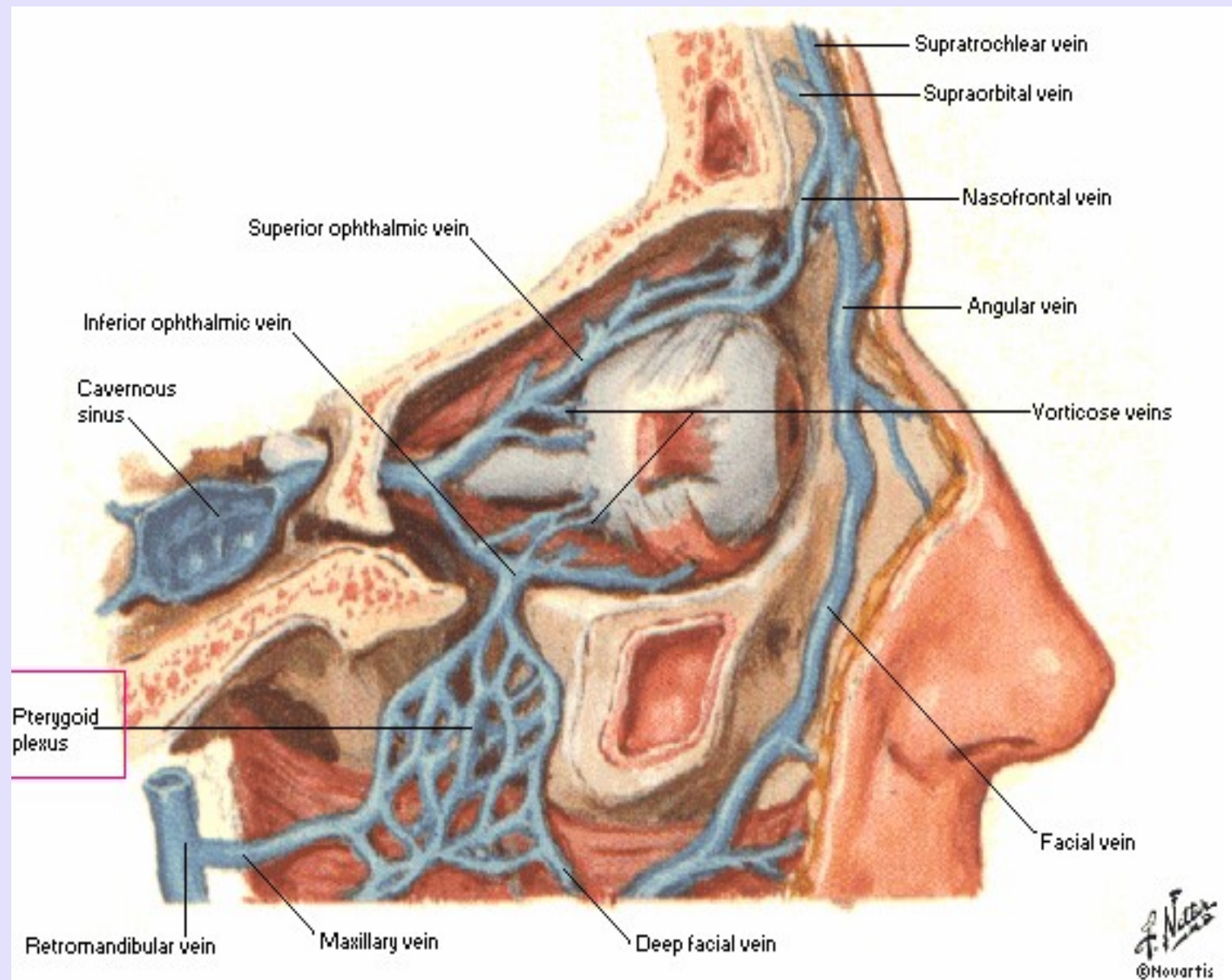
Pharyngeal nerve

Posterior nasal lateral nerves and nasopalat nerve

Pterygopalatine ganglion

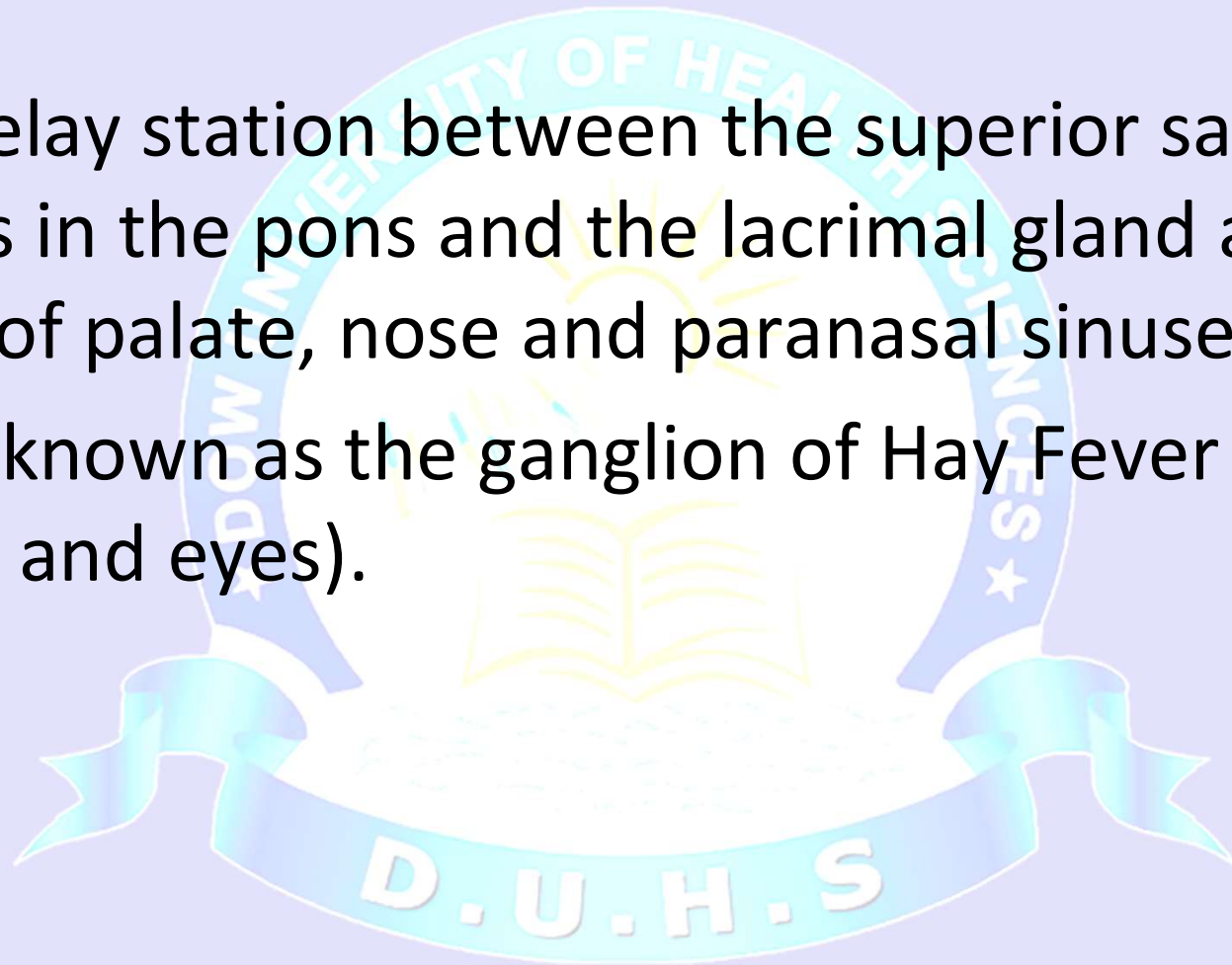
Palatine nerves, greater and lesser



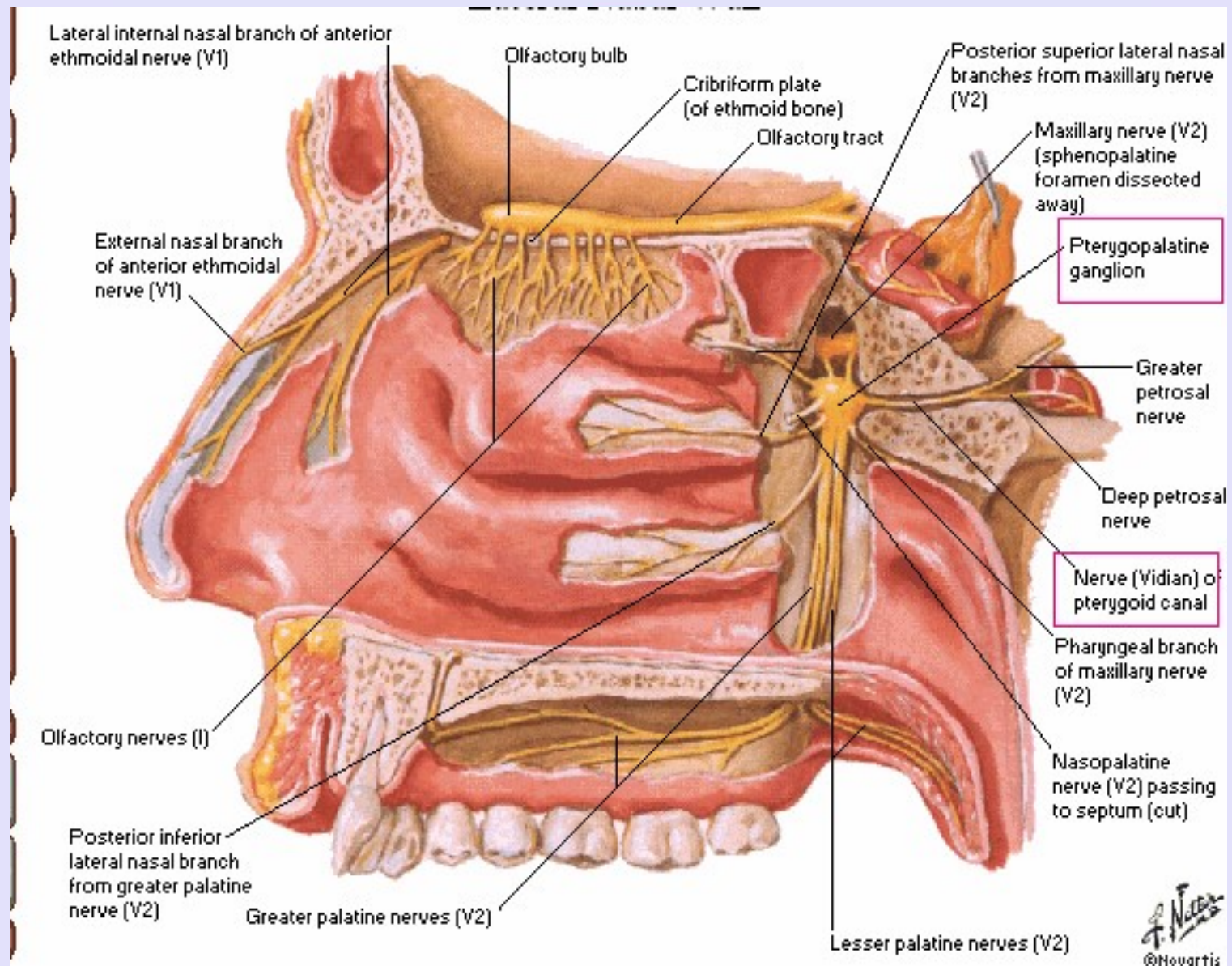


2. PTERYGOPALATINE GLANGLION.

- It is a relay station between the superior salivary nucleus in the pons and the lacrimal gland and the glands of palate, nose and paranasal sinuses.
 - Also known as the ganglion of Hay Fever (running nose and eyes).



Pterygopalatine ganglion.



Pterygopalatine ganglion.

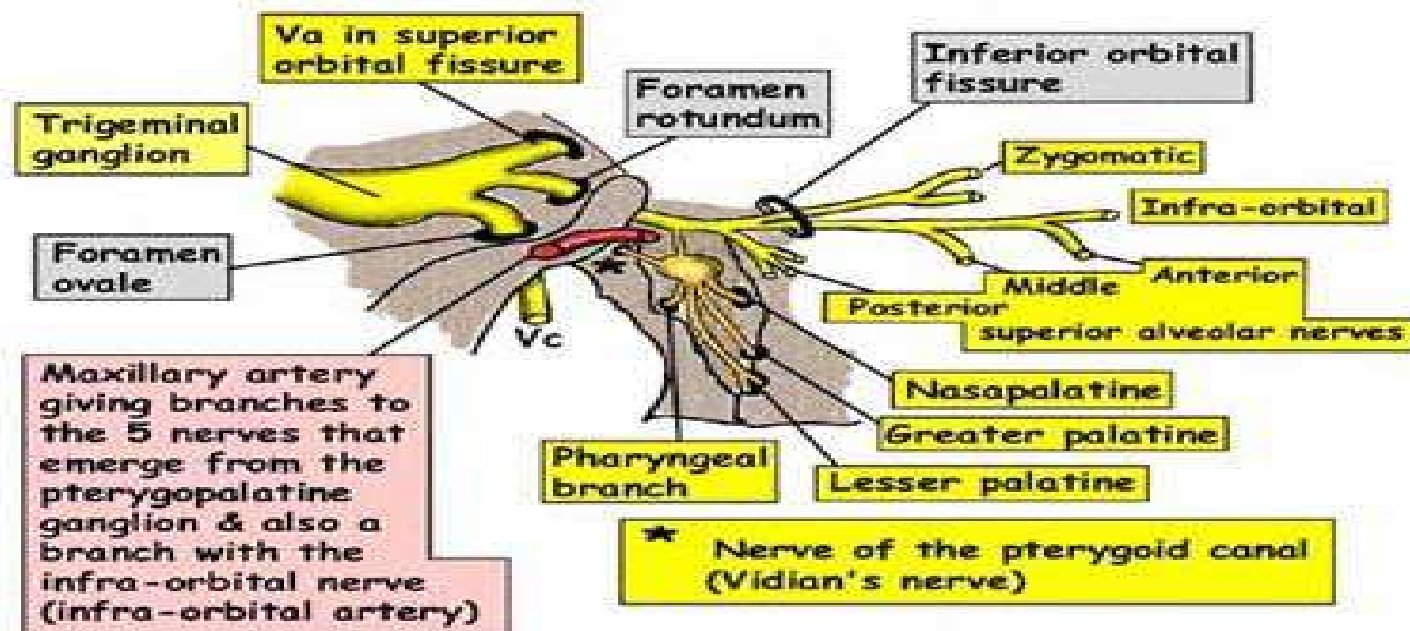
- **Branches:**
- The five branches of pterygopalatine ganglion are distributed to the nose and palate. Every branch carries a mixture of all three kinds of fibers; sensory, parasympathetic and sympathetic.
- **Nasopalatine nerve (Sphenopalatine Nerve)**
Sphenopalatine foramen
→ supplies nasal septum of incisive gum of hard palate
- **Lateral posterior superior nasal nerves (short sphenopalatine nerves)**
Sphenopalatine foramen
→ supplies posterosuperior quadrant of the lateral wall of the nose
- **Greater palatine nerves**
Greater palatine canal greater palatine foramen
supplies hard palate and posteroinferior quadrant of the
→ lateral wall of nose
- **Lesser palatine nerves (two in number)**
Passes behind the greater palatine nerve lesser palatine foramen
→ supplies soft palate and mucous membrane of palatine tonsil
- **Pharyngeal nerve**
Palatovaginal canal emerges at the roof of the nose
supplies the mucous membrane of the upper nasopharynx

PTERYGOPALATINE FOSSA 1

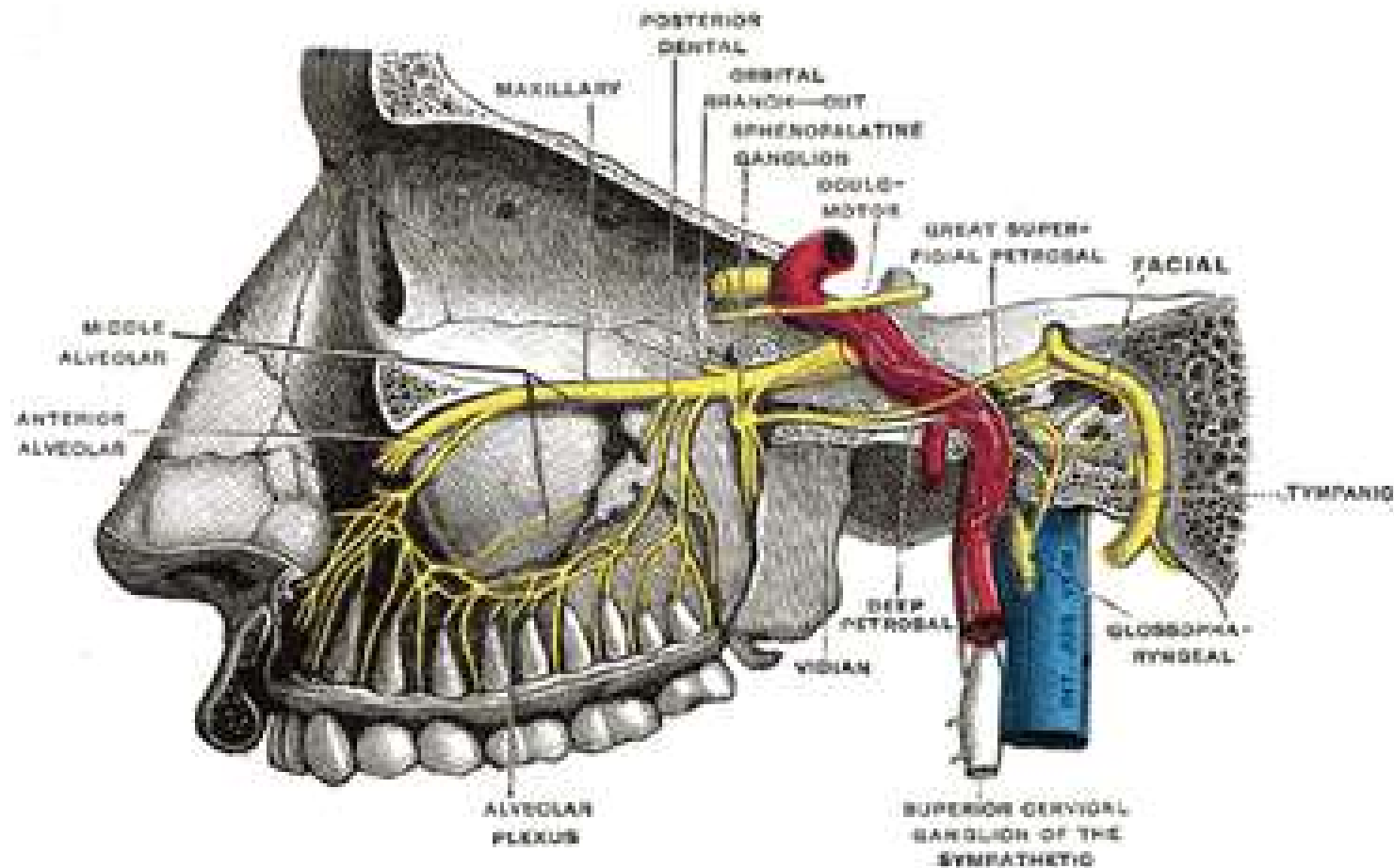
Right side of skull cut away to show trigeminal ganglion lying in Meckel's cave and the maxillary division entering the pterygopalatine fossa through foramen rotundum. The nerve of the pterygoid canal is seen entering the pterygopalatine ganglion and connecting to Vb so that sensory fibres can be distributed with the parasympathetic fibres from the ganglion and so that parasympathetics can pass on Vb to be distributed to sinuses and lacrimal gland.

The contents of the pterygopalatine fossa are:

- Terminal branches of the maxillary artery
- Maxillary nerve (Vb) to upper teeth, floor of orbit, face/skin
- Pterygopalatine ganglion for distribution of parasympathetics to nose and palate



Connections and branches of pterygopalatine ganglion



CONNECTIONS.

Parasympathetic

Lacrimary nucleus of
cranial nerve VII

ganglion



Nervus intermedius



Greater petrosal nerve

Sympathetic

Plexus of nerve around
internal carotid artery



Deep petrosal nerve



Nerve to the Pterygoid canal

Sensory

From a branch of the maxillary nerves,
with the cell bodies in the trigeminal
ganglion. Pass through the
without relay



3. Maxillary nerve:

It is a division of the trigeminal nerve



Passes through the foramen rotundum



Reaches pterygopalatine ganglion



Enters inferior orbital fissure

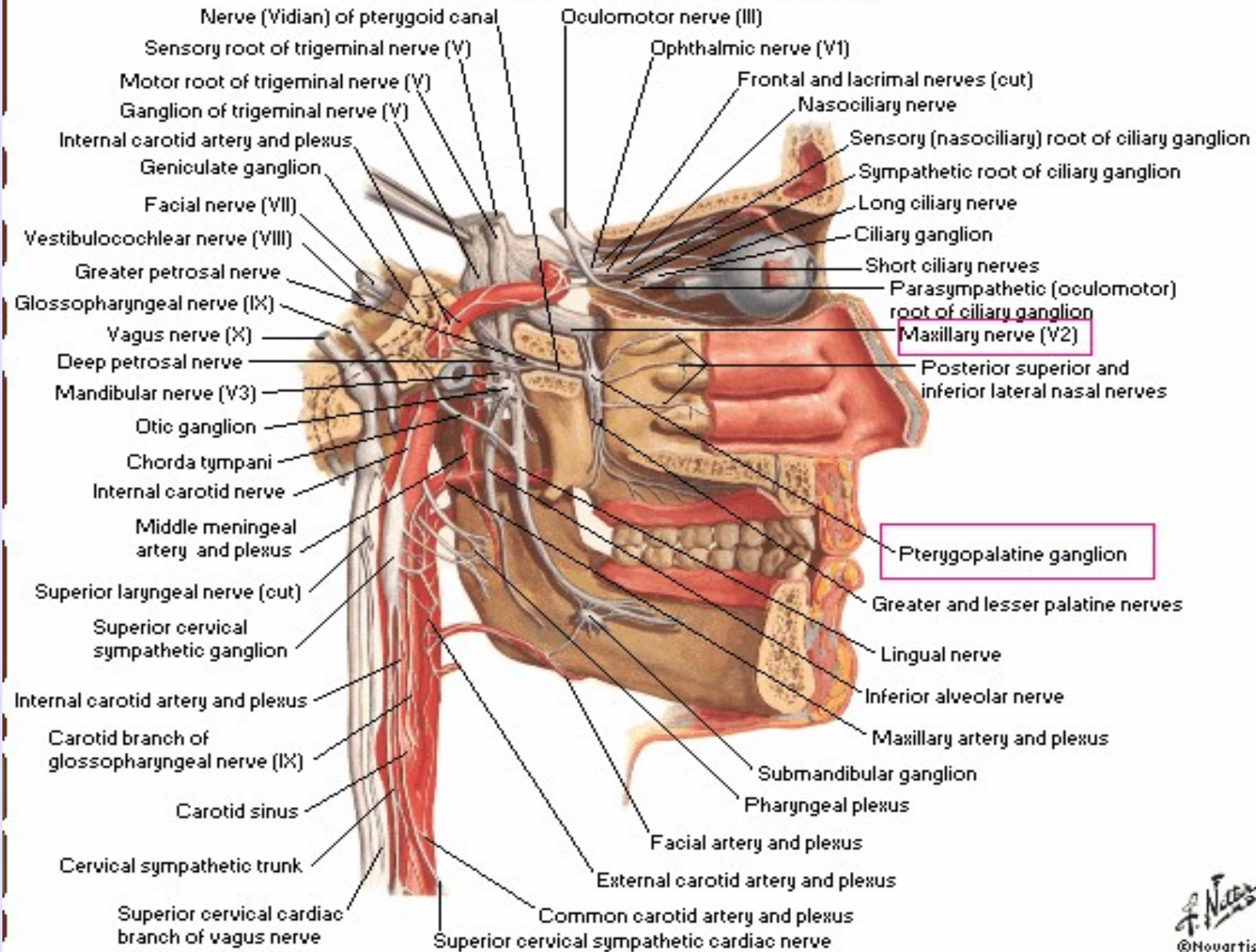


Passes through the inferior orbital groove and canal



Changes its name to infraorbital nerve as it comes out of the infraorbital foramen

Autonomic Nerves in Head



Branches.

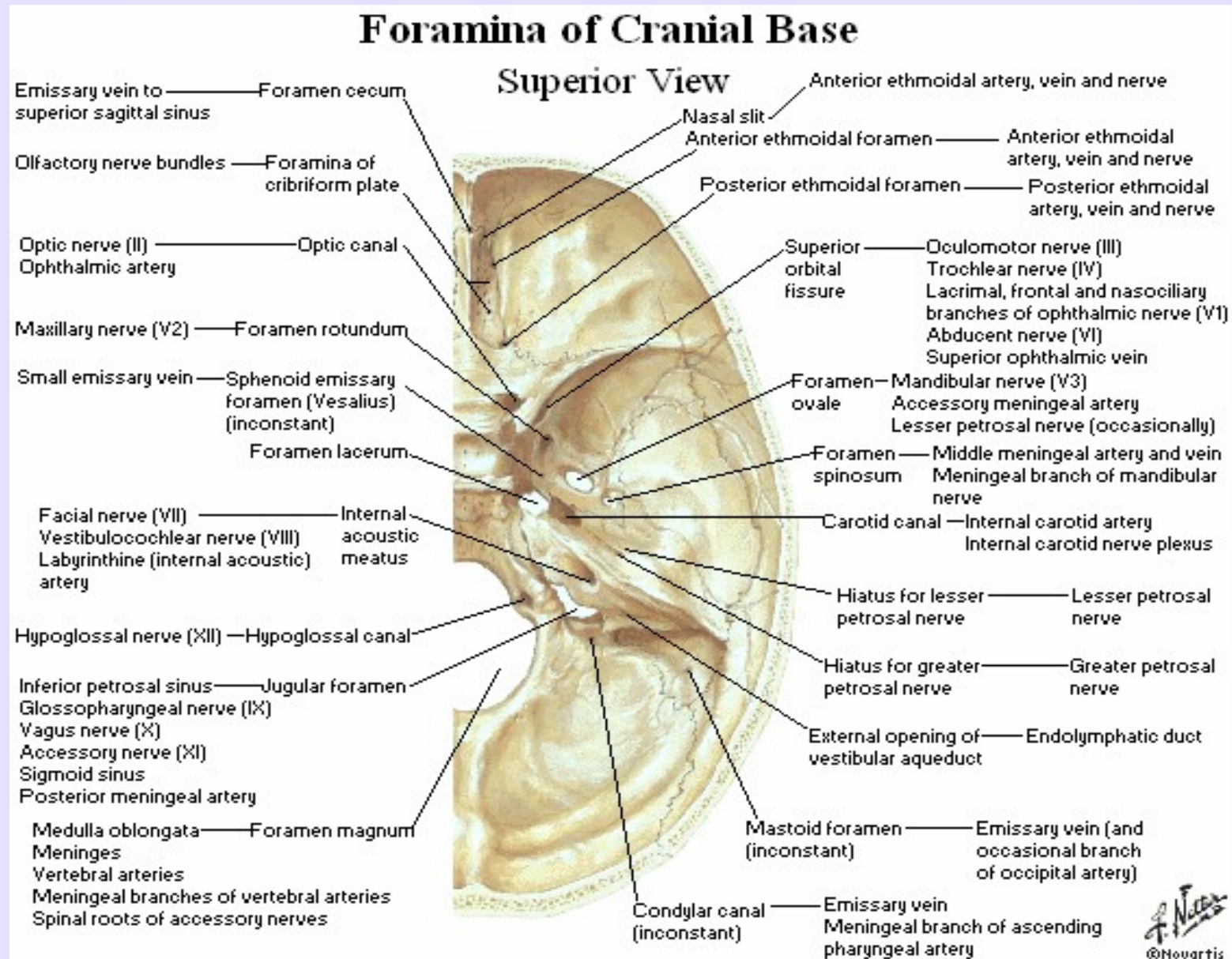
- In the middle cranial fossa:
 - Meningeal branch
- In the Pterygopalatine fossa:
 - Zygomatic nerves → inferior orbital fissure
→ orbit
 - Posterior superior alveolar nerve → Pterygo-maxillary fissure.

Trunk connecting the maxillary nerve to the Pterygopalatine ganglion.

OPENINGS IN PTERYGOPALATINE FOSSA.

Walls	Passage	Contents	Passage communicates with
Lateral	Pterygomaxillary fissure	Maxillary artery Maxillary vein Posterior superior alveolar nerve Branch of maxillary nerve	Infratemporal fossa
Medial	Sphenopalatine foramen	Nasopalatine nerve → Pterygopalatine ganglion Lateral posterior superior nasal nerve Sphenopalatine artery → maxillary artery	Nose
Anterior	Inferior orbital fissure	Maxillary nerve Zygomatic nerve → Branch of maxillary nerve	Orbit
Posterior	Palatovaginal canal	Pharyngeal branch of maxillary artery Pharyngeal branch of pterygopalatine ganglion	Pharynx
	Foramen Rotundum	Maxillary nerve	Middle cranial fossa
	Pterygoid canal	Nerve to pterygoid canal → Pterygopalatine ganglion Artery to Pterygoid canal → maxillary artery	Foramen lacerum
Inferior	Greater palatine foramen	Greater palatine nerve → Pterygopalatine ganglion Greater palatine artery Maxillary artery	Plate (oral cavity)
	Lesser palatine foramen	Lesser palatine nerve → Pterygopalatine ganglion Lesser palatine arteries → Branches of greater palatine artery	Palate (oral cavity)

OPENINGS IN PTERYGOPALATINE FOSSA.



Self Assessment.

1. Pterygopalatine fossa is bounded by which parts of skull bones?
2. What are the contents of pterygopalatine fossa?
3. Name the branches of terminal part of maxillary artery?
4. Pterygopalatine ganglion divides into? (name them).
5. name the openings in pterygopalatine fossa?

